Feature Split–Merge–Enhancement Network for Remote Sensing Object Detection

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***Abstract*— Recently, multicategory object detection in high-resolution remote sensing images is still a challenge. First, objects with significant scale differences exist in one scene simultaneously, so it is generally difficult for the detectors to balance the detection performance of large and small objects. Second, because of the complex background and the objects' densely distributed characteristics in the remote sensing images, the extracted features usually have noise and blurred boundaries, which interfere with the detection performance of the object detectors. With this observation, we propose an end-to-end scale-aware network called feature split–merge–enhancement network (SME-Net) for remote sensing object detection, composed of the feature split-and-merge (FSM) module, the offset-error rectification (OER) module, and the object saliency enhancement (OSE) strategy. FSM eliminates salient information of large objects to highlight the features of small objects in the shallow feature maps. It also transmits the effective detailed features of large objects to the deep feature maps, alleviating feature confusion between multiscale objects. OER corrects the inconsistency of the features spatial layout among the multilayer feature maps by the proposed offset loss, so as to achieve supervised elimination and transmission in FSM. OSE enhances the features of interests and suppresses the background information by the proposed membership function, thus preventing false detection and missed detection caused by noise and blurred boundaries. The effectiveness of the proposed algorithm has been verified on multiple datasets. Our code is available at: https://github.com/Momuli/SMENet.git**

***Index Terms*— Deep learning, feature enhancement, multiscale objects, object detection, remote sensing images.**

I. Introduction

OBJECT detection of remote sensing images [1]–[5] has always been a fundamental and significant task in remote sensing image interpretation. It not only needs to determine the categories of the objects but also needs to calculate the correct position of each object. It plays an important role in both civil and military applications, such as military reconnaissance, resource exploration, disaster prediction, etc. However, quickly and effectively using remote sensing image information and achieving more accurate detection is still a challenge.

With the rise of deep learning, especially in natural images, object detection based on deep learning has made significant progress. It can be mainly divided into two-stage object detectors and single-stage object detectors. Two-stage object detectors [6]–[10] are usually based on the interesting region for detection. They have achieved high detection accuracy, but their speed is usually slow in actual engineering applications. The single-stage detectors [11]–[16] can be regarded as regression analysis models to analyze the relationship between the pixel value of the input image and the variables, such as object position and category information [17], [18]. They transform the object detection task as an end-to-end regression analysis problem of object position and category information, and realize bounding box regression and category prediction through the network models. They have a higher detection speed, but the detection accuracy is less than that of two-stage object detectors.

Most of the object detection methods in remote sensing images are inspired by natural images. However, because of the particularity of remote sensing imaging technology, there are still many challenges under the complex backgrounds in actual scenes. The following challenges are obvious.

1) Remote sensing images often contain small objects submerged by complex surrounding scenes, such as ships in Fig. 1(a) and cars in Fig. 1(b).
2) The objects are densely distributed in the same scene, such as storage tanks in Fig. 1(c) and harbors in Fig. 1(d).

For the former challenge, in order to improve the detection accuracy of small objects, multiscale object detection [19]–[25] and feature fusion object detection [26]–[31] are widely studied.

For the multiscale object detection, Ding *et al.* [19] presents a multiscale representation and various combinations of improvement schemes to enhance the backbone for improving precision. Wang *et al.* [20] proposed a novel one-stage detector called the full-scale object detection network (FSoD-Net), which consists of the proposed multiscale enhancement network backbone cascaded with scale-invariant regression layers (SIRLs). Zhang *et al.* [21] proposes a semantic context-aware network (SCANet) model for multiscale object detection to extract multiscale feature representations and sufficiently exploit semantic context information. However, shallow fea-

Manuscript received July 16, 2021; revised October 12, 2021 and December 6, 2021; accepted December 20, 2021. Date of publication January 6, 2022; date of current version March 8, 2022. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant JB211909 and Grant XJS201904; in part by the National Natural Science Foundation of China under Grant 62006179; in part by the China Postdoctoral Science Foundation funded project under Grant 2019M663634 and Grant 2020T130492; in part by the Key Scientific Technological Innovation Research Project by the Ministry of Education; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-05; and in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61621005. *(Corresponding author: Hao Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3140856

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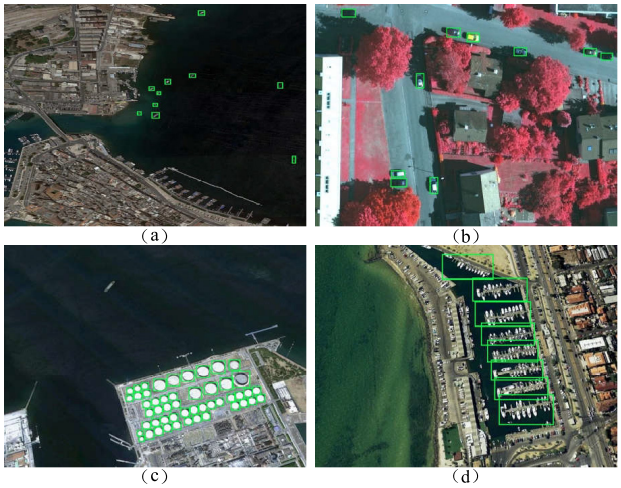


Fig. 1. Two main challenges of object detection in remote sensing images. Remote sensing images often contain small objects submerged by complex surrounding scenes, such as (a) ships and (b) cars. The objects are densely distributed in the same scene, such as (c) storage tanks and (d) harbors.

ture maps are mainly responsible for detecting small objects, while detailed information of the large and small objects exist concurrently. The large objects usually dominate the main saliency and weaken the small objects' feature, thus hindering the effective detection of the small objects.

For the object detection based on feature fusion, Wang *et al.* [26] proposed a unified framework called feature-merged single-shot detection (FMSSD) network to aggregate the context information in multiscale and the same scale feature maps. Teng *et al.* [27] proposed an innovative global to local object detection network (GLNet) for remote sensing imagery integrates global context clues and local spatial contextual correlations encoded by clip long short-term memory (Clip-LSTM). Lu *et al.* [28] proposes an end-to-end network named attention and feature fusion single-shot multibox detector (SSD), which designs a multilayer feature fusion structure to enhance the semantic information of the shallow features. Nevertheless, feature fusion between multilayer feature maps may lead to more severe confusion in the scale of features, and it also introduces disturbing information into the feature maps.

For the latter challenge, aimed at intensive objects, Guo *et al.* [32] proposes a rotational Libra R-convolutional neural network to balance the three levels (feature level, sample level, objective level) of neural networks for predicting the location of objects with rotational angle information. Wang *et al.* [33] proposed a weighted pseudo-segmentation-guided attention network to distinguish better the intensive objects from the cluttered background. Yang *et al.* [34] proposes the supervised pixel attention network and the channel attention network to detect densely distributed objects. Although the rotated bounding box can usually detect the dense objects better, nevertheless, it needs to predict the angle difference between the anchor and ground truth, and the deviation of angle regression results has a great impact on the final detection results.

In this article, we do not merely show solicitude for the small objects but pay attention to the large objects' scale difference caused by the coexistence of large and small objects in the same scene.

For the large-scale difference of objects, we propose the *feature split-and-merge* (FSM) module; it eliminates the features of the large objects in the shallow feature maps and mainly retains the small objects' features, so the network can better capture the small objects, improving the ability to detect small objects. It also suppresses the salient features of large objects in the shallow layer, reducing false-positive problems. In addition, the significant detailed information of the large objects is transferred to the deep feature maps, which makes up for the insufficiency of the detailed feature in the deep feature maps while improving the overall detection performance. We also introduce an *offset-error rectification* (OER) module to guide the elimination and transmission process.

For small and densely distributed objects, considering the blurred boundary and noise in the feature maps, we propose an *object saliency enhancement* (OSE) strategy. It enhances the object characteristics and suppresses background noise with purpose according to the established membership function.

Our contributions and innovations are summarized as follows.

1) For the large objects' scale difference, we propose an *FSM* module in the network mechanism. It eliminates salient features of large objects to emphasize the features of small objects in the shallow feature maps. It also transmits detailed information of large objects to the deep feature maps to enhance feature aggregation; thus, the large and small objects in one scene are distributed in the multilevel feature maps for subsequent detection, alleviating feature confusion between multiscale objects.
2) The *OER* module is proposed to correct the inconsistency of objects' spatial layout among the multilayer feature maps, so as to achieve supervised irrelevant feature elimination and effective detailed information transmission (DIT) in the FSM module.
3) For the complex background and dense objects, we propose an *OSE* strategy, which separates the foreground and background by the proposed membership function. It enhances the features of interests, suppresses the features of no interests, and decreases false detection and missed detection caused by noise and blurred boundaries.

The rest of this article is organized as follows. Section II introduces the work related to the proposed method. Section III elaborates the proposed method in detail. Section IV introduces the details of datasets used in this article and the experimental setup, then shows the experimental results and the corresponding analysis. Finally, Section V concludes this article.

II. Related Work

In this section, two works related to our algorithm will be introduced in detail, including the multiscale object detection algorithms and the feature enhancement algorithms in object detection.

A. Multiscale Object Detection

Large-scale variation [35] across object instances has always been a very challenging problem in object detection, especially

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in remote sensing images. Therefore, to improve the detection accuracy of different scale objects, some multiscale object detection algorithms [11], [36]–[42] have gradually appeared.

Among them, the SSD [11] uses multilayer feature maps for detection. It combines the regression idea of YOLO [12] and the anchor mechanism of faster regions with CNN features (R-CNN) [8], then realizes the predictive regression of multiscale feature maps. However, the semantic information of the shallow feature maps is insufficient, resulting in unsatisfactory results for small object detection. The deconvolutional single-shot detector [36] augments SSD with deconvolution layers to introduce additional large-scale context. The feature fusion SSD [37] makes features from different layers concatenate together, and then uses some down-sampling blocks to generate a new feature pyramid that will be fed to multibox detectors to predict the final detection results. Multiscale object detection in remote sensing imagery with convolutional neural networks [39] proposes network [a multi-scale object proposal network (MS-OPN)] for object-like region generation from several intermediate layers, whose receptive fields match different object scales, and also proposes an accurate object detection network (AODN) for object detection based on fused feature maps, which combines several feature maps that enable small and densely packed objects to produce stronger response. SSD with Rainbow Concatenation (RSSD) [41] uses pooling and deconvolution to change the resolution of feature maps, realizes feature concatenation between multilayer feature maps, and makes the classifier network fully utilize the relationship between the layers in the feature pyramid. FPN [42] exploits the inherent multiscale, pyramidal hierarchy of deep convolutional networks to construct feature pyramids, and a top–down architecture with lateral connections is developed for building high-level semantic feature maps at all scales. However, the straightforward fusion methods between multilayer feature maps usually lead to more serious confusion [43] in the scale of features, and even aggravate the occurrence of false negatives and false positives.

Moreover, the training objective of the above methods usually divided into two parts, including object categories and object locations. Therefore, the overall objective loss function is a weighted sum of the localization loss and the confidence loss. It can be expressed as

$$L(x, c, l, g) = \frac{1}{N}(L_{\text{conf}}(x, c) + \beta L_{\text{loc}}(x, l, g)) \tag{1}$$

where x is an indicator that reflects whether the default box matches the ground truth. The localization loss (L_{loc}) is a smooth L1 loss between the predicted box (l) and the ground-truth box (g) parameters. The confidence loss (L_{conf}) is the softmax loss over multiple classes confidences (c). The weight term β is set to 1 by cross validation. Specific definitions can be found in literature [11].

However, the loss function equation (1) treats each sample loss equally and pays the same attention to the hard and easy samples. Actually, to improve the discriminative ability of the model, the network should pay more attention to the hard samples. In addition, for the problem of sample imbalance between categories, when there are more numbers of category c_1 and fewer category c_2, the network parameters will be dominated by the loss of category c_1, which greatly reduces the detection accuracy of category c_2.

B. Feature Enhancement in Object Detection

The ability of feature representation is directly related to the accuracy of the subsequent detection results. Considering the complex background and the dense distribution characteristics of objects in remote sensing images, a large amount of noise information may be introduced in feature extraction and feature fusion stage. It makes the boundary between dense objects fuzzy, and weakens ability of feature representation, increasing the possibility of missing or false detection. To obtain high-quality feature maps, feature enhancement [44]–[48] and noise reduction can be performed on the feature maps. It can enhance the overall or local characteristics of the images, emphasize some features of interests, expand the differences between different object features in the feature maps, and suppress the features of no interests.

Traditional image enhancement methods are mainly divided into two parts. The former is the spatial domain methods, which directly process the pixel values, such as histogram equalization [49]–[51] and gamma transform [52], [53]. The latter is the frequency domain methods, which operate in a certain transform domain, such as wavelet transform [54]. These methods are generally simple and fast, but little consideration is given to the context information in the image.

In the field of deep learning, the multiscale interactive network for salient object detection [55] proposes an enhancement loss function to help the model highlight the whole salient region, then better deal with the pixel imbalance between the front and back regions. Basnet [56] introduced structural similarity loss, which focuses on solving boundary ambiguity, because under the loss of structural similarity, the loss of boundary will be more weighted than that given by the interior or other places of the significant objects. Snider [57] uses U-Net [58] to enhance the detailed information of the object in the images without having a negative impact on image generation. At the same time, it adds skip connection to share the low-level semantic information of the images to realize feature enhancement. These methods usually emphasize objects features by fusing multilevel and multiscale feature information, which will inevitably amplify or introduce noise from challenging samples.

III. METHOD

The framework of the proposed method is summarized in Fig. 2. For the input image, we first employ the backbone to get multilevel feature maps. Subsequently, the FSM module and the OSE strategy are adopted to get more effective feature maps; the OER module corrects the inconsistency of objects spatial layout among the multilevel feature maps in the networks' training process. Finally, different detection heads are used for box regression and classification. We present the details of the FSM module, the OER module, and the OSE strategy as follows.

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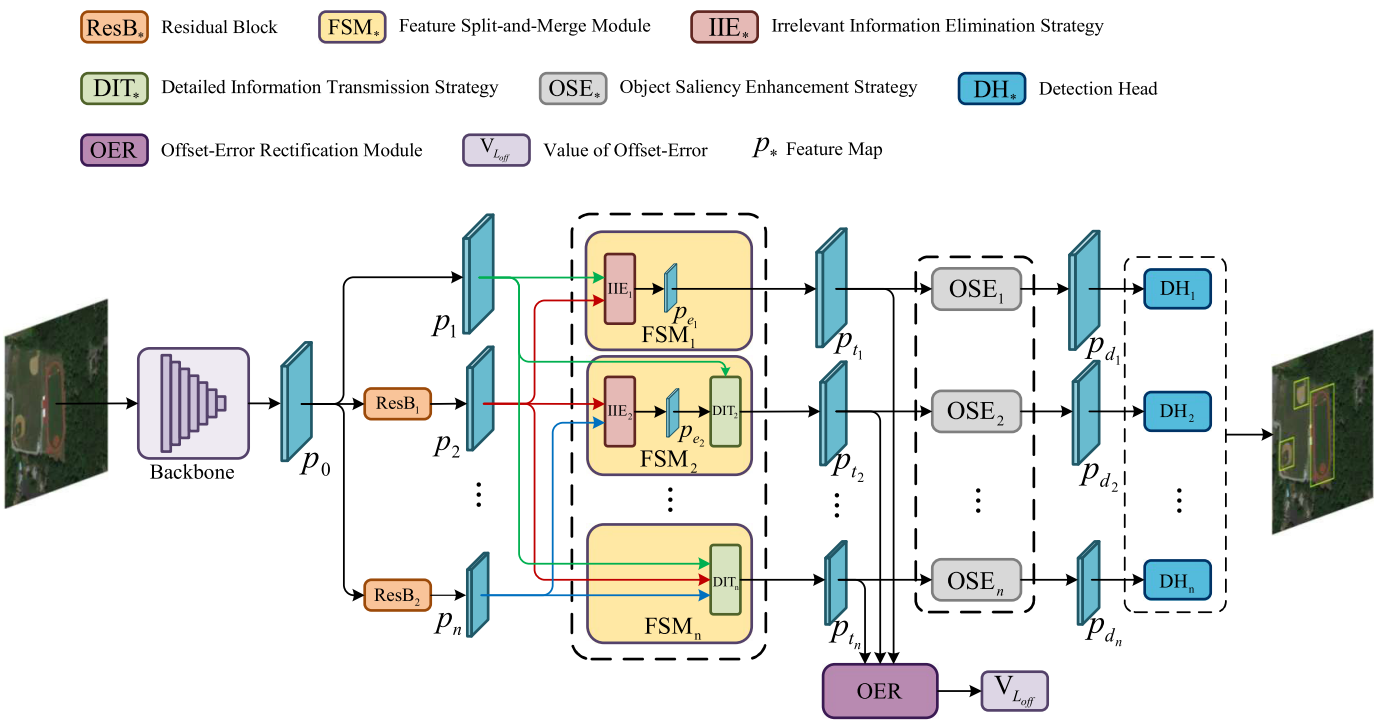


Fig. 2. Framework of split–merge–enhancement network (SME-Net). First, the input image is processed by the backbone and multiple residual blocks to obtain multilevel feature maps ($p_1, p_2, \ldots, p_n$). And then, FSM modules are adopted to eliminate interference information and transfer effective features in multilevel feature maps. Subsequently, the feature maps are, respectively, fed into OSE strategies to reinforce salient features. Furthermore, the OER module is used to correct the inconsistency of objects' spatial layout among the multilevel feature maps in the network's training process. Finally, box regression and classification are completed by different detection heads. In particular, p_1 is the shallowest feature map, so FSM_1 does not contain the DIT strategy for transmitting detailed information. p_n is the deepest feature map, so FSM_n does not contain the irrelevant information elimination (IIE) strategy for eliminating irrelevant features. In our experiments, we obtain feature maps p_1, p_2, and p_3 for subsequent processing, so $n = 3$. In Section III, we illustrate the proposed FSM module, OER module, and OSE strategy in detail.

A. FSM Module

Usually, both large and small objects exist in one remote sensing image simultaneously. Because of the significant difference in the objects' scale, the features of the small objects are easily lost during extracting feature information as the plain network deepens. Therefore, for object detection of remote sensing images, object detectors similar to SSD are often used to complete independent detection of multiscale feature maps, i.e., the shallow feature maps are used to detect small objects, while the deep feature maps are used to detect large objects.

However, the detailed features of large and small objects exist in the shallow feature maps concurrently, as shown in Fig. 3(a), while the large objects' information tends to be more salient. We find that the object detectors are more inclined to pay more attention to the salient feature information through experiments, so it is still difficult to detect small objects in the shallow feature maps. In addition, the partial regions of the large objects usually correspond to strong responses, which also lead to the false-positive detection of large objects.

Furthermore, both shallow and deep feature maps have different tendencies of feature expression. The shallow feature maps are rich in detailed features but lack semantic features. On the contrary, the deep feature maps have strong semantic feature representation capabilities but lose many detailed features. Traditional object detection algorithms often apply various feature fusion strategies such as top-down feature pyramid [36], [37], [42], yet they also introduce disturbing information into the feature maps. This usually leads to feature confusion among multiscale objects; it means that feature

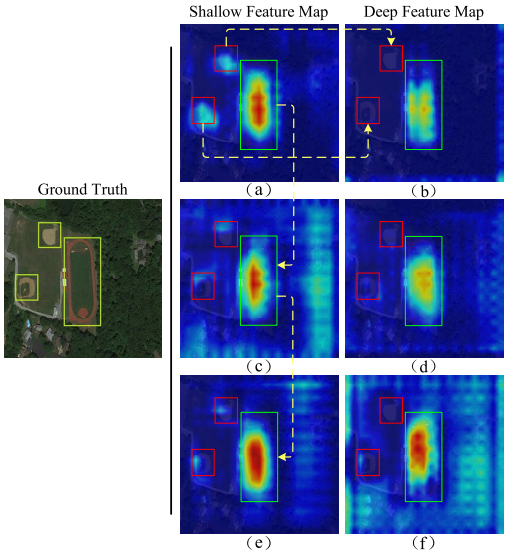


Fig. 3. Visualized feature maps are extracted from the adjacent layer of the backbone as the number of iterations increases during the training process. Horizontal observation [e.g., (a) and (b); (c) and (d); and (e) and (f)] shows that information about small objects is gradually lost as the network gets deeper. Vertical observation [e.g., (a), (c), and (e) or (b), (d), and (f)] shows that the network pays more attention to the salient regions of large objects gradually. We use Grad-CAM [59] to visualize the feature maps.

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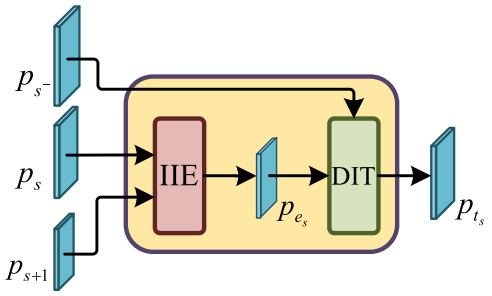


Fig. 4. FSM module. It consists of the IIE strategy and the DIT strategy. The adjacent feature maps p_s and p_{s+1} generate p_e without irrelevant information by the IIE strategy. p_e obtains effective detailed information from the shallow feature maps p_{s^-} and generates p_{t_s} by the DIT strategy.

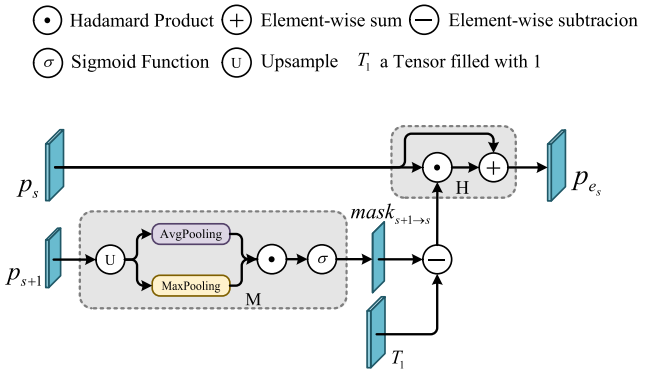


Fig. 5. IIE strategy.

information of some large objects will be introduced into the shallow high-resolution feature maps, which are originally responsible for detecting small objects, making feature information of the small objects not pure enough, thereby reducing the discriminative ability of the small objects' features, deteriorating the performance of detection.

In response to the above problems, this article proposes an *FSM* module, as shown in Fig. 4. It consists of the *IIE* strategy and the *DIT* strategy.

To obtain multilevel feature maps for subsequent operations, for the input image $I \in \mathbb{R}^{W\times H\times 3}$, $p_0 \in \mathbb{R}^{W_0\times H_0\times C_0}$ is the feature map extracted from feature extractor. We get multilevel feature maps $\{p_1, p_2, \ldots, p_n\}$ from p_0 processed by multiple residual blocks. It can be expressed as

$$p_i = C_{\text{rb}}[p_0, s_d = R_i], \quad (i = 1, 2, \ldots, n) \tag{2}$$

where $C_{\text{rb}}[p_0, s_d = R_i]$ represents the feature map $p_i \in \mathbb{R}^{(W_0/R_i)\times(H_0/R_i)\times C_i}$ is obtained by $(R_i/2)$ residual blocks, and the down-sampling step is R_i. In this article, we set $n = 3$. We aim to detect small objects with the shallow feature maps and large objects with the deep feature maps.

1) IIE Strategy: Fig. 3 shows adjacent hierarchical feature maps $p_s \in \mathbb{R}^{W_s\times H_s\times C_s}$ [see Fig. 3(a)] and $p_{s+1} \in \mathbb{R}^{W_{s+1}\times H_{s+1}\times C_{s+1}}$ [see Fig. 3(b)], which are extracted from the backbone. We note $X = \{x_1, x_2, \ldots, x_n\}$ as objects set for all scales, where x_1 represents objects with the smallest scale and x_n represents objects with the largest scale.

According to experiments, we find that p_s is more inclined to detect objects similar to the size of x_s, but p_s contains feature information of multiscale objects $\{x_s, x_{s+1}, \ldots, x_n\}$ [60], [61]. While p_{s+1} mainly contains the feature information of large objects $\{x_{s+1}, x_{s+2}, \ldots, x_n\}$, feature information of the small objects x_s is almost lost.

Since the small objects have less visualization information, as the network deepens, the position and feature information of the small objects are gradually lost. Therefore, it is difficult to extract the discriminative features of small objects from the deep feature maps.

Based on the above features spatial layout, in the *IIE* strategy, as shown in Fig. 5, we obtain $\text{mask}_{s+1\to s} \in [0, 1]^{W_s\times H_s\times 1}$ from deeper feature map p_{s+1}, which contains the stronger response value on the regions of objects $\{x_{s+1}, x_{s+2}, \ldots, x_n\}$. It can be expressed as

$$\begin{aligned}\text{mask}_{s+1\to s} &= M(p_{s+1}, s_u = 2)\\ &= \sigma\left(P_{\text{avg}}(U(p_{s+1})) \odot P_{\max}(U(p_{s+1}))\right)\end{aligned} \tag{3}$$

where $M(\cdot)$ represents a function to obtain a spatial mask. s_u in $M(\cdot)$ represents upsamples rate. Function $U(p_{s+1})$ upsamples p_{s+1} to p_s to keep a consistent spatial resolution. $P_{\text{avg}}(\cdot)$ and $P_{\max}(\cdot)$, respectively, represent *Avg_pooling* and *Max_pooling* along channels to get spatial layout characteristics of objects $\{x_{s+1}, x_{s+2}, \ldots, x_n\}$ in the feature map p_{s+1}. $\odot$ refers to *Hadamard product*, and $\sigma(\cdot)$ means *Sigmoid* function.

We adopt $\text{mask}_{s+1\to s}$ to generate a soft reverse gate for suppressing the features of larger objects $\{x_{s+1}, x_{s+2}, \ldots, x_n\}$ in p_s. Finally, we obtain the feature map $p_{e_s} \in \mathbb{R}^{W_s\times H_s\times C_s}$ that mainly contains features of objects x_s by function $H(\cdot)$. To avoid gradient vanishing in the network training process, function $H(\cdot)$ can be expressed as

$$\begin{aligned}p_{e_s} &= H(1 - \text{mask}_{s+1\to s}, p_s)\\ &= p_s \odot (1 - \text{mask}_{s+1\to s}) + p_s.\end{aligned} \tag{4}$$

2) DIT Strategy: The feature map p_{e_s} obtained by *IIE* mainly contains feature information of objects x_s. However, taking p_{e_s} as a reference, the imbalance between the semantic information and detailed information in the feature map still exists, which means the detailed information of larger objects in the deep feature maps is still insufficient. In this article, the *DIT* strategy, as shown in Fig. 6, is adopted to enhance the representation ability of detailed information for the deep feature maps.

Specifically, we define a *transmission unit* (TU) as an essential component for DIT, as shown in Fig. 6. In *TU*, for the feature map p_{e_s} that is obtained from p_s by *IIE*, and the feature maps p_{s^-} (i.e., $p_{s-1}, p_{s-2}, \ldots, p_1$) that shallower than p_s, similar to *IIE*, we get $\text{mask}_{s\to s^-} \in \mathbb{R}^{W_{s^-}\times H_{s^-}\times 1}$ as follows:

$$\text{mask}_{s\to s^-} = M\left(p_{e_s}, s_u = \frac{W_{s^-}}{W_s}\right) \tag{5}$$

$\text{mask}_{s\to s^-}$ has stronger activation values on the regions of objects x_s, so we use $\text{mask}_{s\to s^-}$ as a guide to extract pure detailed information of objects x_s from p_{s^-}. Consequently, we get $p_{s\to s^-} \in \mathbb{R}^{W_{s^-}\times H_{s^-}\times 1}$ as follows:

$$p_{s\to s^-} = p_{s^-} \odot \text{mask}_{s\to s^-} \tag{6}$$

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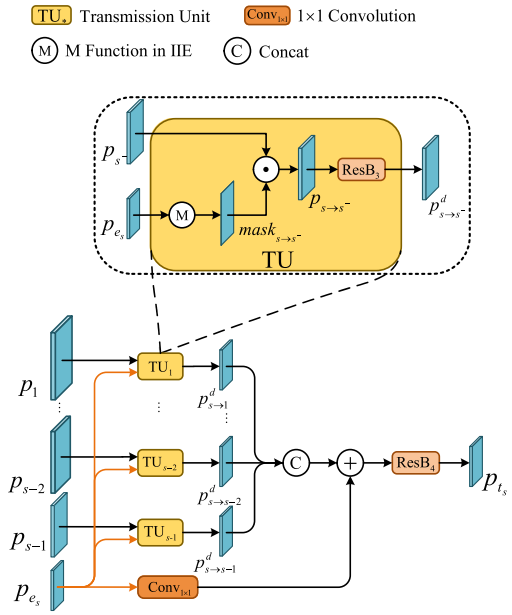


Fig. 6. DIT strategy. TU is a component for DIT. p_{e_s} adopts TU to obtain effective detailed information from the shallow feature maps (e.g., p_1, p_{s-2}, and p_{s-1}).

$p_{s\to s^-}$ contains useful detailed information of objects x_s. We use residual blocks to keep the consistent spatial resolution between $p_{s\to s^-}$ and p_{e_s}. Thus, we get $p^d_{s\to s^-} \in \mathbb{R}^{W_s\times H_s\times C_s}$ in the following way:

$$p^d_{s\to s^-} = C_{\text{rb}}\left[p_{s\to s^-}, s_d = \frac{W_{s^-}}{W_s}\right]. \tag{7}$$

Finally, in the DIT strategy, we transmit the useful detailed features of objects x_s contained in $p^d_{s\to s^-}$ to the feature map p_{e_s} as follows:

$$p_{t_s} = C_{\text{rb}}\left[C_{1\times 1}(p_{e_s}) + Cat\left(p^d_{s\to s-1}, p^d_{s\to s-2}, \ldots, p^d_{s\to 1}\right), s_d = 1\right] \tag{8}$$

where Cat(·) represents concatenation operation. It aggregates all the detailed feature information of objects x_s, which are contained in the feature maps that are shallower than the feature map p_s, so as to enhance the detailed information representation ability of the feature map p_s. $C_{1\times 1}(\cdot)$ refers to $1 \times 1 \times C_s$ convolution layer. It compresses the channels of feature map p_{e_s} to ensure that it can fuse with the feature map obtained by Cat(·) operation. The feature maps $p^d_{s\to s-1}$ and $p^d_{s\to s-2}, \ldots, p^d_{s\to 1}$ are obtained from (7). $p_{t_s} \in \mathbb{R}^{W_s\times H_s\times C_s}$ is the feature map that contains the shallow useful detailed features of objects x_s.

To avoid introducing redundant noise information caused by the mutual fusion between multi-layer features, we replace it with multiple residual blocks to directly enhance the semantic representation abilities of feature maps.

The FSM module divides objects into smaller scale intervals for subsequent detection, alleviating feature confusion between multiscale objects. Specifically, the *IIE* strategy eliminates salient features of the large objects, then mainly retains the features of small objects in the shallow feature maps so that the network can better capture the small objects. It also reduces the occurrence of false positive detection of large objects. In addition, the pure detailed information of large objects in the shallow feature maps is transmitted to the deep feature maps through the DIT strategy, which makes up for the defect of insufficient detailed information in the deep feature maps.

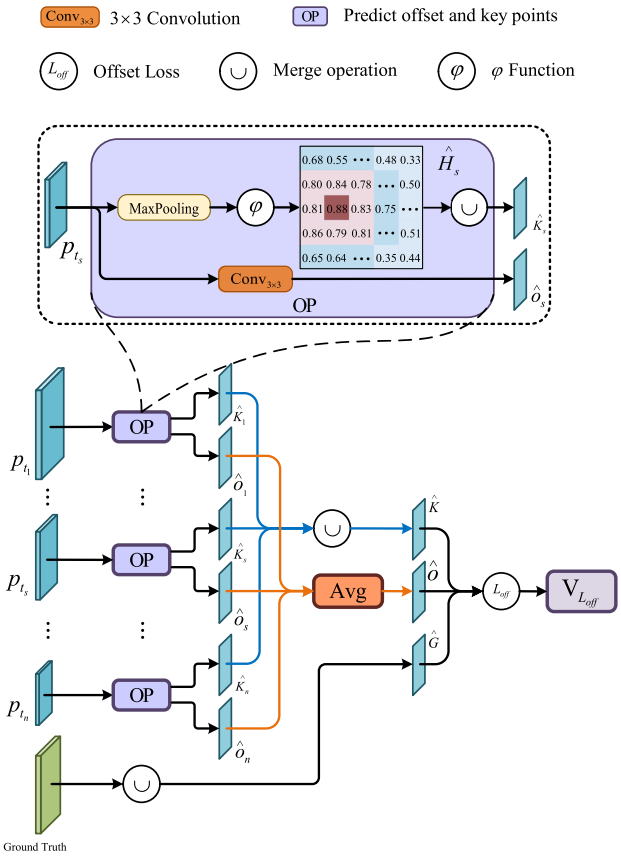


Fig. 7. OER module. $\hat{H}_s$ refers to a heatmap; we adopt it to obtain key points set.

B. OER Module

In the FSM module, to realize the information interaction between multilevel feature maps, we adopt upsample operation on the feature map p_{e_s} to keep the consistent spatial resolution with the shallower feature maps p_{s^-}, as shown in (5). Because of the inconsistency of spatial information between adjacent feature maps, straightforward upsample operation may lead to eliminate favorable feature or transmit interference noise. To suppress the phenomenon, this article proposes an *OER* module, as shown in Fig. 7.

Specifically, for the feature map p_{t_s} obtained from FSM, we set $\hat{O}_s \in \mathbb{R}^2$ as the predicted offset-error and $\hat{K}_s \in \mathbb{R}^2$ as the key points set. It can be expressed as

$$\hat{O}_s = \left(\Delta\hat{x}_s, \Delta\hat{y}_s\right) \tag{9}$$

$$\hat{K}_s = \{(a_i, b_i)\}_{i=1}^{n_s} \tag{10}$$

where $\Delta\hat{x}_s$ and $\Delta\hat{y}_s$ represent the offset in the horizontal and vertical directions, respectively. (a_i, b_i) refers to the coordinate of the key point k_i. n_s represents the number of key points in the feature map p_{t_s}.

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In the OER module, the predicted offset-error $\hat{O}_s$ is calculated by 3×3 convolution with learnable weights w_s. It can be expressed as follows:

$$\hat{O}_s = C_{3\times3}\left(p_{t_s}, w_s\right). \tag{11}$$

To locate the regions of learned objects features in feature map p_{t_s}, we first introduce the *Max_pooling* operation along the channel to focus on more meaningful position features. And then, we find the possible key points that have a high probability of being in the objects' regions through the heatmap $\hat{H}_s \in \mathbb{R}^{W_s \times H_s \times 1}$, hence we obtain heatmap $\hat{H}_s$ as follows:

$$\begin{aligned}\hat{H}_s &= \varphi\left(P_{\max}\left(p_{t_s}\right)\right) \\ &= \frac{P_{\max}\left(p_{t_s}\right)}{\text{max-min}}\end{aligned} \tag{12}$$

where max and min refer to the maximum and minimum values in the feature map p_{t_s}.

For the heatmap $\hat{H}_s$, we set $A(a, b)$ as a key point if the value of A is greater than or equal to its eight connected neighbors, then record its coordinate. $\hat{K}_s$ is the set of n_s key points coordinates obtained from heatmap $\hat{H}_s$.

We generate the total set of key points $\hat{K} \in \mathbb{R}^2$ as follows:

$$\hat{K} = \bigcup_s \left(\hat{K}_s\right) \tag{13}$$

where $\bigcup(\cdot)$ represents to merge the key points of multiple feature maps. Considering the complexity of the model, all objects share the same offset-error; thus, the total offset-error $\hat{O} \in \mathbb{R}^2$ is obtained by averaging the predicted offset-error $\hat{O}_s$ of each feature map as follows:

$$\hat{O} = \text{Avg}\left(\sum_s \hat{O}_{s\,\Delta x}, \sum_s \hat{O}_{s\,\Delta y}\right). \tag{14}$$

In addition, we get the center points set $\hat{G} \in \mathbb{R}^2$ according to ground truth, and it can be expressed as

$$\hat{G} = \bigcup_{j=1}^{N} \left(x_j, y_j\right) \tag{15}$$

where N refers to the total number of objects in the ground truth, and (x, y) represents the coordinate of objects.

The final offset loss (L_{off}) is defined as follows:

$$L_{\text{reg}}\left(\hat{K}, \hat{G}\right) = \left(\left|\sum_{i=1}^{n} \hat{K}_{a_i} - \sum_{j=1}^{N} \hat{G}_{x_j}\right|, \left|\sum_{i=1}^{n} \hat{K}_{b_i} - \sum_{j=1}^{N} \hat{G}_{y_j}\right|\right) \tag{16}$$

$$\begin{aligned}L_{\text{off}} &= \frac{1}{n} \times L_d \times L_v = \frac{1}{n} \times \frac{L_{\text{reg}}\left(\hat{K}, \hat{G}\right)}{\left|L_{\text{reg}}\left(\hat{K}, \hat{G}\right)\right|} \\ &\quad \times \left[\left(\hat{O}_{\Delta x} - \left|\sum_{i=1}^{n} \hat{K}_{a_i} - \sum_{j=1}^{N} \hat{G}_{x_j}\right|\right)\right. \\ &\quad \left. + \left(\hat{O}_{\Delta y} - \left|\sum_{i=1}^{n} \hat{k}_{b_i} - \sum_{j=1}^{N} \hat{G}_{y_j}\right|\right)\right]\end{aligned} \tag{17}$$

where $L_{\text{reg}}(\hat{K}, \hat{G})$ represents position deviation calculated in horizontal and vertical directions. L_d determines the direction of gradient propagation, and L_v refers to the magnitude of the gradient.

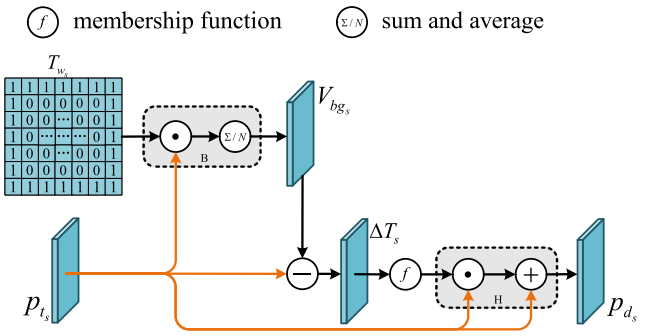


Fig. 8. OSE strategy. T_{w_s} is a learnable weight tensor. We initialize the edge values of T_{w_s} as 1 and the values of other regions as 0.

To supervise the learning of offset-error, we adopt key points set $\hat{K}$ and center point set $\hat{G}$ as a pseudo label to make the predicted offset-error more accurate. It can be seen that when the value of L_v tends to zero, the model will be more robust to the situation of spatial information inconsistency between multilayer feature maps, thereby effectively reducing the interference of this inconsistency in detection performance.

The total loss (Loss) is a weighted sum of the localization loss (L_{loc}), the confidence loss (L_{conf}), and the offset loss (L_{off}). It can be expressed as

$$\text{Loss} = \frac{1}{N}\left(L_{\text{conf}}(x, c) + \beta L_{\text{loc}}(x, l, g)\right) + \gamma L_{\text{off}}\left(\hat{K}, \hat{G}, \hat{O}\right). \tag{18}$$

Similar to SSD, x is an indicator that reflects whether the default box matches the ground truth. The localization loss (L_{loc}) is a smooth L1 loss between the predicted box (l) and the ground-truth box (g) parameters. The confidence loss (L_{conf}) is the softmax loss over multiple classes confidences (c). The hyperparameters β, γ control the tradeoff.

The OER module corrects the inconsistency of the features spatial layout among the multilayer feature maps and avoids eliminating beneficial information and introducing interference noise, so as to achieve supervised elimination of irrelevant features and transmission of useful, detailed information.

C. OSE Strategy

Because of the particularity of remote sensing imaging technology, the natural factors, such as cloud, mist, and illumination usually disturb and occlude objects, making the objects' features incomplete. Thus, it leads to miss objects in detection. In addition, because of the complex background and the dense distribution characteristics of remote sensing images, it is possible to introduce a lot of noise information in the process of feature extraction. Noise may blur the boundary between dense objects and weaken the ability of objects' feature representation.

Based on the above analysis, this article proposes an *OSE* strategy, as shown in Fig. 8. In general, the edge pixels of the scene are considered as the background [62]. For the feature map p_{t_s}, we set $V_{bg_s} \in \mathbb{R}^{W_s \times H_s \times C_s}$ as the representative value

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of the background. We calculate V_{bg_s} as follows:

$$V_{bg_s} = \mathrm{Avg}\big(\Sigma\left(T_{w_s} \odot p_{t_s}\right)\big) \tag{19}$$

where $T_{w_s} \in \mathbb{R}^{W_s \times H_s \times C_s}$ is a learnable weight tensor; we initialize the edge values of T_{w_s} as 1 and the values of other regions as 0. We use T_{w_s} to get all edge pixels of the feature map p_{t_s}, and set the average value of all edge pixels to represent the background V_{bg_s}. In the training process of the network, the elements' values of T_{w_s} are continuously optimized, so as to make the calculated average value from the feature maps better represent the background. Then, we regard $\Delta T_s \in \mathbb{R}^{W_s \times H_s \times C_s}$ as the difference between the feature map p_{t_s} and V_{bg_s}. It can be calculated as follows:

$$\Delta T_s = p_{t_s} - V_{bg_s}. \tag{20}$$

Generally, there is a large difference between the elements' value corresponding to the regions of objects and the background. Therefore, for the element $\Delta T_{s_{(i,j)}}$, the larger its value is, the corresponding location in feature map p_{t_s} is more likely to contain salient information, and a higher attention should be paid to the region. To describe the positive correlation, the membership function $f(\cdot)$ is defined as follows:

$$f\left(\Delta T_{s_{(i,j)}}\right) = \begin{cases} 1 - e^{-\alpha \times \Delta T_{s_{(i,j)}}}, & \Delta T_{s_{(i,j)}} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

where α is a hyperparameter, and we analyze it in Section IV.

Then a saliency map is obtained by $f(\Delta T_s)$. It should be noted that the value of $f(\Delta T_s)$ is between [0, 1], and it reflects the membership degree to which each element in the feature map p_{t_s} belongs to the foreground. In other words, it relatively emphasizes the regions of interest and suppresses regions of no interest. So, the final feature map $p_{d_s} \in \mathbb{R}^{W_s \times H_s \times C}$ for detection is determined as follows:

$$p_{d_s} = p_{t_s} \oplus \left(p_{t_s} \odot f(\Delta T_s)\right). \tag{22}$$

The OSE strategy separates the foreground and background according to the membership function; it enhances the features of objects and reduces the noise, thus preventing false detection and missed detection caused by noise and blurred boundaries.

IV. Experiments

In this section, we first introduce the datasets used in our experiments. Next, the implementation details and the hyperparameter analysis are provided. Later, we prove the effectiveness of each proposed module through a series of experiments. Finally, the detection results and comparisons with state-of-the-art methods are presented.

A. Dataset Description

In this article, we use two datasets to validate our proposed method: one is NWPU VHR-10 [63] and the other is IPIU-M.

The NWPU VHR-10 dataset is a challenging ten-class object-detection dataset annotated by Northwestern Polytechnical University. This dataset contains 800 images, of which 650 contain objects, and the remaining 150 contain no objects.

TABLE I
Different Category Object Numbers in NWPU VHR-10

	number	Mean Resized Scale /pixel	Resized Scale Std /pixel
airplane	757	67.67	19.39
baseball diamond	390	84.26	22.95
basketball court	159	93.80	33.79
bridge	124	152.74	45.80
ground track field	163	275.19	51.81
harbor	224	99.35	23.54
ship	302	52.45	14.83
storage tank	655	39.57	9.47
tennis court	524	62.60	15.50
vehicle	477	41.74	9.61

TABLE II
Different Category Object Numbers in IPIU-M

	number	Mean Resized Scale /pixel	Resized Scale Std /pixel
helicopter	727	19.19	3.98
Boeing	672	19.89	4.09
dense_ship	18697	19.24	4.05
sparse_ship	4417	23.44	5.51

Of these images, 715 are high-resolution remote sensing images collected from Google Maps with spatial resolutions ranging from 0.5 to 2 m, and the remaining 85 images are pan-sharpened color-infrared images with a spatial resolution of 0.08 m. Table I summarizes the object numbers of all the categories. The mean resized scale column and the resized scale standard column, respectively, indicate the average and the standard deviation of all the objects in a specific category. In this dataset, the *vehicles* are usually in complex scenes and have a smaller scale, and the *storage tanks* are usually densely distributed. Therefore, they are more challenging to detect than other categories.

Another dataset is IPIU-M, which contains 1739 images acquired from Google Earth and Baidu Map with a spatial resolution from 0.15 to 1.2 m. There are 24 513 object instances in IPIU-M, and they are divided into four categories. The object numbers and the object scales of different categories are listed in Table II. Small objects (e.g., *helicopter*) and dense objects (e.g., *dense_ship*) occupy a larger proportion in this dataset.

In our experiments, we randomly select half the dataset as training data and the other half as test dataset.

B. Experimental Setup

The feature extractor of our model is based on the successful VGG16 [64], pretrained on the ImageNet dataset [65]. The convolutional layer and pooling layer of VGG16 can be divided into several blocks (Block), which are sequentially

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TABLE III
CONFIGURATIONS OF OUR NETWORK

Module	Input	Filter Size	Stride	Downsampling	Output
$VGG16_{[1-4]}$	$S \times S, 3$	—	8	Yes	$S/8 \times S/8, 512$
$Preprocess$	$S/8 \times S/8, 512$	$ResB[512, 1024, 2]$ $ResB[512, 512, 2]$ $ResB[512, 1024, 2]$	2	Yes	$S/16 \times S/16, 1024$
			4	Yes	$S/32 \times S/32, 1024$
IIE_1	$S/8 \times S/8, 512$ $S/16 \times S/16, 1024$	$Upsample_mask[1024, 2]$	1	No	$S/8 \times S/8, 512$
IIE_2	$S/16 \times S/16, 1024$ $S/32 \times S/32, 1024$	$Upsample_mask[1024, 2]$	1	No	$S/16 \times S/16, 1024$
DIT_2	$S/8 \times S/8, 512$ $S/16 \times S/16, 1024$	TU_1: $Upsample_mask[1024, 2]$ $ResB[512, 512, 2]$	2	Yes	$S/16 \times S/16, 512$
	$S/16 \times S/16, 1024$	$Conv: 1 \times 1, 1024 \rightarrow 512$ $ResB[512, 512, 1]$	1	No	
DIT_3	$S/8 \times S/8, 512$ $S/32 \times S/32, 1024$	TU_1: $Upsample_mask[1024, 4]$ $ResB[512, 512, 4]$	4	Yes	$S/32 \times S/32, 512$
	$S/16 \times S/16, 1024$ $S/32 \times S/32, 1024$	TU_2: $Upsample_mask[1024, 2]$ $ResB[1024, 1024, 2]$	2	Yes	
	$S/32 \times S/32, 1024$	$Conv: 1 \times 1, 1024 \rightarrow 1536$ $ResB[1536, 512, 1]$	1	No	
OER	$S/8 \times S/8, 512$	$Conv: 3 \times 3, 512 \rightarrow 2$	1	No	$S/8 \times S/8, 2$
	$S/16 \times S/16, 512$	$Conv: 3 \times 3, 512 \rightarrow 2$	1	No	$S/16 \times S/16, 2$
	$S/32 \times S/32, 512$	$Conv: 3 \times 3, 512 \rightarrow 2$	1	No	$S/32 \times S/32, 2$
$Classification$	$S/8 \times S/8, 512$	$Conv: 3 \times 3, 512 \rightarrow classes \times k$ $(unshared)$	1	No	$S/8 \times S/8, classes \times k$
	$S/16 \times S/16, 512$				$S/16 \times S/16, classes \times k$
	$S/32 \times S/32, 512$				$S/32 \times S/32, classes \times k$
$Localization$	$S/8 \times S/8, 512$	$Conv: 3 \times 3, 512 \rightarrow 4 \times k$ $(unshared)$	1	No	$S/8 \times S/8, 4 \times k$
	$S/16 \times S/16, 512$				$S/16 \times S/16, 4 \times k$
	$S/32 \times S/32, 512$				$S/32 \times S/32, 4 \times k$
$ResB[In, Out, S]$	$W \times H, C$	$Conv: 1 \times 1, In \rightarrow In/2$ $Conv: 3 \times 3, In/2 \rightarrow In/2$ $Conv: 1 \times 1, In/2 \rightarrow Out$	$S = 1$	No	$W \times H, out$
			$S = 2$	Yes	$W/2 \times H/2, out$
$Upsample_mask[In, S]$	$W \times H, C$	$Upsample: Bilinear$ $Avg_Pooling_{Channel}$ $Max_Pooling_{Channel}$ $Sigmoid()$	1	No	$S * W \times S * H, 1$

numbered Block1–Block5. Each block contains several convolutional layers and a pooling layer. In our experiments, we only keep Block1–Block4 of VGG16, and we extract feature map p from the output of the final convolution result of Block4 for subsequent processing. The configurations of our network are summarized in Table III.

The training hyperparameters are summarized in Table IV. To capture objects of various sizes, we define the anchor sizes of 20, 55, 130, and the ratios of 0.65, 1, 2. They correspond to feature maps sent to the detection heads with the sizes of 50, 25, and 13 pixels, respectively. In addition, to avoid image compression, we resize the input images to 400×400 pixels.

During the test period, the *nonmaximum suppression* (NMS) method is used to reduce the redundant boxes. The *NMS* first selects the box with the highest score among all the boxes. Then, it compares other boxes with the selected box and

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TABLE IV
TRAINING PARAMETERS

parameters	values
Input Image Size	400 pixel
Anchor Size	[20, 55, 130]
Ratios	[0.65, 1, 2]
Weight Initilization	Xaiver
Learning Rate	0.0001
Momentum	0.9
Weight Decay	0.0005
Max Iteration	28500

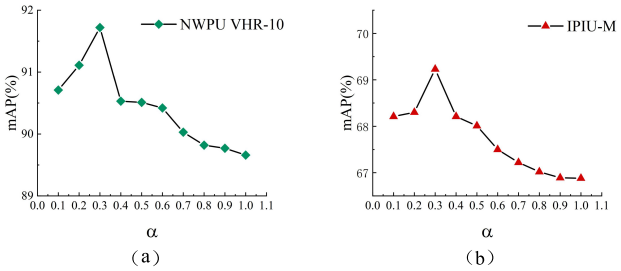


Fig. 9. Detection performance for different values of α on NWPU VHR-10 and IPIU-M. (a) NWPU VHR-10. (b) IPIU-M.

removes boxes if their intersection-over-union (*IOU*) with the selected box is greater than the *NMS* threshold. Finally, these two steps are repeated among those unselected boxes until all the input boxes are selected or removed. The *NMS* threshold is 0.45 in our experiments.

The experiments in this article are running on workstation with RTX2080Ti 12 GB GPU and 128 GB RAM under Ubuntu 16.04 LTS. The proposed network is trained on PyTorch.

We do all experiments five times and take the average value as the final result to avoid the influence of abnormal situations.

C. Comparison and Analysis of Hyperparameters

This section provides a detailed comparison and analysis of three hyperparameters in our network: α in membership function equation (21) and the weights γ and the weights β in total loss equation (18). We adopt the average precision (AP) to evaluate the object detection performance.

Here, except for the hyperparameters that need to be determined, each dataset is trained and tested with the same other parameters. We use the fixed variable method to carry out the experiment. We optimize the hyperparameters on the NWPU VHR-10 dataset and IPIU-M dataset, respectively.

1) Hyperparameter Analysis for α: The range of membership function equation (21) is [0, 1]; the hyperparameter α determines the change speed of membership function from 0 to 1 and indirectly affects the probability that elements in the feature maps belong to the foreground. Here, we only add the OSE strategy to the baseline. To be specific, we vary the value of the α from 0 to 1 with a stride of 0.1 and provide the corresponding results in Fig. 9.

As can be seen, in both datasets, the best detection performance is achieved when $\alpha = 0.3$, while the performance

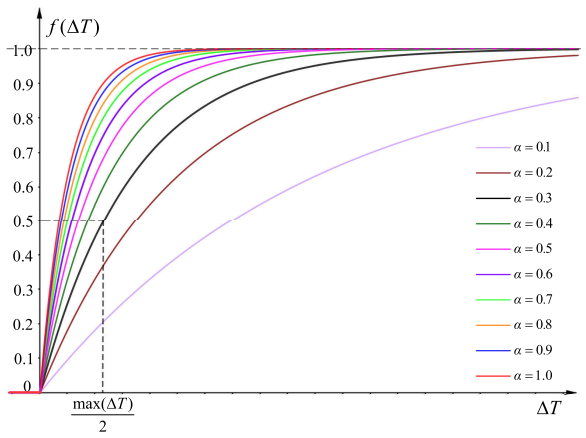


Fig. 10. Membership function with different values of α.

TABLE V
EXPERIMENTAL RESULTS OF HYPERPARAMETER γ AND β ON NWPU VHR-10 (%)

hyper-paramers		mAP
SSD-VGG16 ∪ FSM		93.21
$\beta = 0.5$	$\gamma = 0.1$	92.47
$\beta = 0.5$	$\gamma = 0.3$	93.96
$\beta = 0.5$	$\gamma = 0.5$	92.92
$\beta = 0.4$	$\gamma = 0.3$	92.85
$\beta = 0.3$	$\gamma = 0.3$	92.72
$\beta = 0.2$	$\gamma = 0.3$	92.54

TABLE VI
EXPERIMENTAL RESULTS OF HYPERPARAMETER γ AND β ON IPIU-M (%)

hyper-paramers		mAP
SSD-VGG16 ∪ FSM		69.31
$\beta = 0.5$	$\gamma = 0.1$	70.66
$\beta = 0.5$	$\gamma = 0.3$	70.01
$\beta = 0.5$	$\gamma = 0.5$	69.37
$\beta = 0.4$	$\gamma = 0.3$	68.50
$\beta = 0.3$	$\gamma = 0.3$	68.00
$\beta = 0.2$	$\gamma = 0.3$	66.13

decreases greatly when the value of α is far away from 0.3. Fig. 10 shows the membership function corresponding to the change of α, and it can be seen that with the change of α from 0.1 to 1, the speed of the membership function value from 0 to 1 is gradually accelerated. If the change is too fast or too slow, the prediction probability of a certain element in the feature maps is too high or too low, leading to a large deviation. With the stable convergence of the network, we find that when $\alpha = 0.3$, the membership values of the elements near $(\max(\Delta T)/2)$ are around 0.5, which is in line with the original intention of our model. Therefore, we set the value of α as 0.3.

2) Hyperparameter Analysis for γ and β: We evaluate the effectiveness of total loss equation (18) with different hyperparameters γ and β. We add OER module based on FSM module. It can be seen in Tables V and VI that our model with different parameters consistently improves over the baseline.

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TABLE VII
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON NWPU VHR-10 (%)

	airplane	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle	mAp
SSD-VGG16	90.72	77.34	79.85	94.43	82.60	77.68	90.05	84.27	61.76	62.78	80.15
SSD-VGG16 ∪ SENet	99.14	88.87	87.01	96.97	94.74	94.05	97.54	89.41	88.20	73.27	90.92
SSD-VGG16 ∪ OSE	98.77	90.03	88.78	95.96	96.10	94.43	97.48	91.03	88.26	76.33	91.72
SSD-VGG16 ∪ FSM	95.75	92.33	89.01	97.90	96.31	95.73	100	91.00	94.05	80.06	93.21
SSD-VGG16 ∪ FSM ∪ OER	96.06	93.06	89.84	98.16	97.08	96.99	99.79	91.03	96.73	80.86	**93.96**

TABLE VIII
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON IPIU-M (%)

	helicopter	Boeing	dense_ship	sparse_ship	mAP
SSD-VGG16	61.89	72.89	51.37	71.14	64.32
SSD-VGG16 ∪ SENet	64.05	75.20	54.98	73.81	67.01
SSD-VGG16 ∪ OSE	66.82	76.56	57.17	76.36	69.23
SSD-VGG16 ∪ FSM	69.57	77.58	56.43	73.65	69.31
SSD-VGG16 ∪ FSM ∪ OER	71.50	78.91	57.96	74.37	**70.69**

In the NWPU VHR-10 dataset, the best performance is achieved when the total loss with $\beta = 0.5$ and $\gamma = 0.3$. While in the IPIU-M dataset, the best performance is achieved when $\beta = 0.5$ and $\gamma = 0.1$.

The reason for the different values of γ in the two datasets is that in IPIU-M, the small and intensive objects account for a large proportion. Thus, there are more objects in the same scene. Note that if there is offset loss in (17), then more key points lead to larger offset loss value. This makes the model pay more attention to the offset error in the learning process, which affects the performance of the model. Therefore, compared to the NWPU VHR-10 dataset, in IPIU-M, the smaller weight γ makes the model get a better balance between the classification error and position error.

D. Module Comparison Experiments

In the module comparison experiments, we take both the NWPU VHR-10 and the IPIU-M datasets to verify the effectiveness of our proposed module, including the FSM module, the OER module, and the OSE strategy. We add each module to the baseline network VGG16 in order. Different category APs between our method and VGG16 on the datasets are compared in Tables VII and VIII. Some qualitative detection results are shown in Figs. 11 and 12, with the first column representing SME-Net detections and the right column representing the backbone detections.

1) Validation of the OSE Strategy: In this part, to verify the availability of our proposed OSE strategy, we add squeeze-and-excitation network (SENet) [66] and OSE to VGG16, respectively. For the fairness of the experiment, except for the SENet and OSE, the parameters of other parts are consistent. In Tables VII and VIII, with the first row representing VGG16 detections and the second and third rows representing VGG16 ∪ SENet and VGG16 ∪ OSE detections, respectively.

In Table VII, OSE improves mAp by 0.8% compared with VGG16 ∪ SENet on NWPU VHR-10. This phenomenon is apparent on *vehicle* and *tennis court*.

Taking *vehicle* as an example, as shown in Fig. 13, we visualize the detected feature map from VGG16 ∪ SENet and VGG16 ∪ OSE. It can be seen that compared with the feature map generated from VGG16 ∪ SENet [see Fig. 13(a)], the feature map processed by OSE strategy [see Fig. 13(b)] can not only better emphasize the objects' features but also effectively suppress the interference information. Similarly, in Table VIII, our OSE strategy also improves the mAP in all categories. The OSE strategy fully considers the contrast between the objects' features and the background when processing the feature maps; therefore, it is more targeted for object enhancement and background suppression.

2) Validation of the FSM and OER Module: Similar to the above part, we add the FSM module and the OER module sequentially on the baseline VGG16. The fourth and fifth rows of Tables VII and VIII show the mAP of VGG16 ∪ FSM and VGG16 ∪ FSM ∪ OER, respectively.

It can be seen that our proposed strategy consistently improves over the baseline. And Fig. 14 shows the extracted featured maps after the FSM module operation. It shows that after FSM operation, the objects are distributed in a small detection space according to the scale. Fig. 14(a) only contains smaller objects and Fig. 14(b) contains larger objects.

However, Fig. 15(a) shows that the response regions of the objects may have some position deviation compared with the actual position due to the inconsistency of spatial information between adjacent feature maps. The feature map on Fig. 15(b) is operated by the OER module and the deviation between the corresponding area of the objects and the actual position has been significantly reduced, verifying the effectiveness of our proposed module.

E. Comparison With Other Methods

To quantitatively evaluate the performance of our proposed method, we compare it with several existing methods, including R-Libra R-CNN [32], gated axis-concentrated localization network (GACL) [67], feature aggregation and enhancement network (FAE-Net) [68], Merge-Net [69], multiscale convolutional neural network (MSCNN) [70], structure-guided feature transform hybrid residual network (SGFTHR) [71], and multiscale semantic fusion-guided fractal convolutional object

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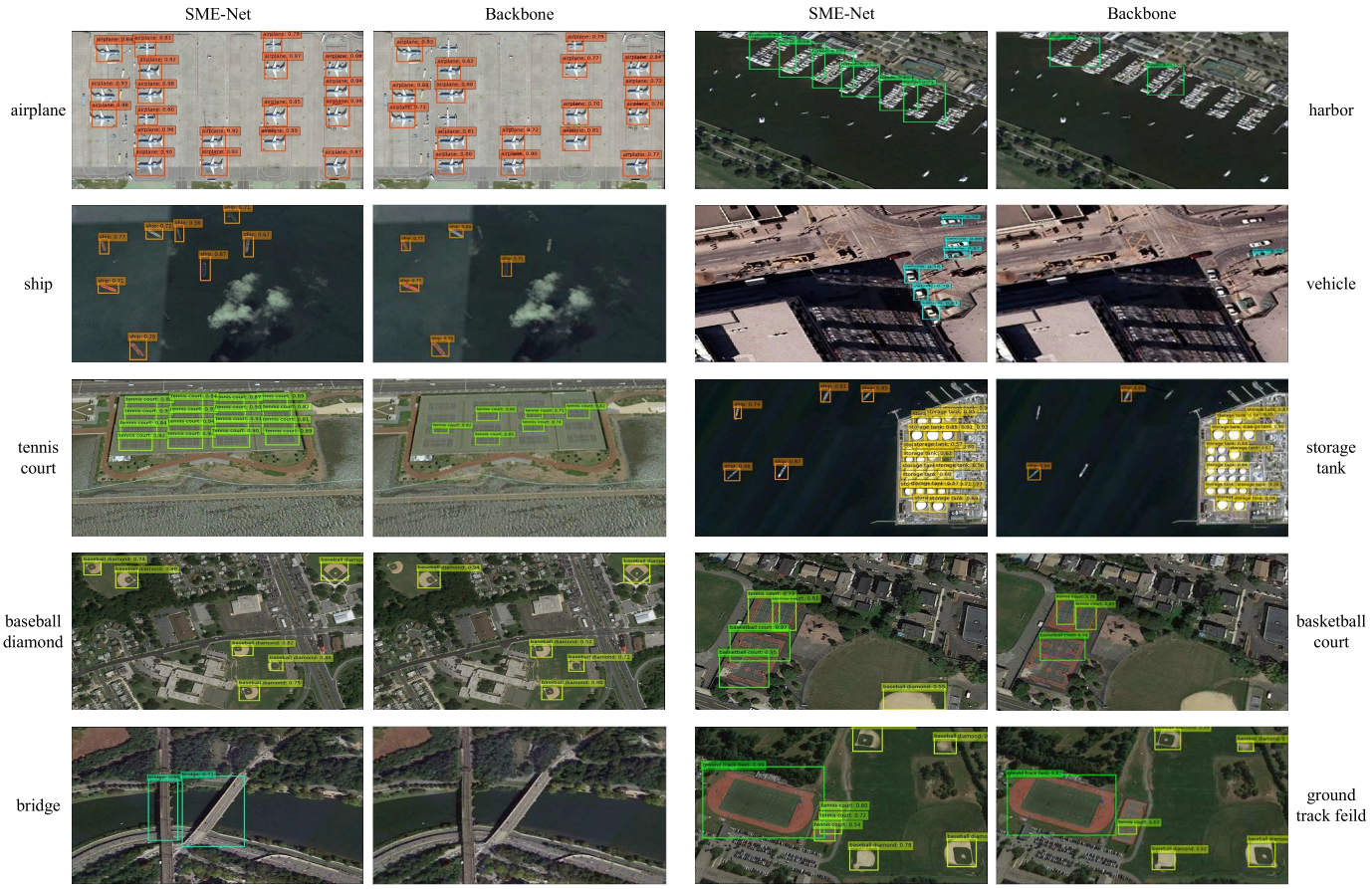


Fig. 11. Detection results comparison between the backbone network and the proposed SME-Net on NWPU VHR-10. Numbers on those boxes are their confidence scores.

TABLE IX
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON NWPU VHR-10 (%)

	airplane	ship	storage tank	baseball diamond	tennis court	basketball court	ground track field	harbor	bridge	vehicle	mAP
SSD-VGG16	90.72	77.34	79.85	94.43	82.60	77.68	90.05	84.27	61.76	62.78	80.15
R-Libra R-CNN	97.09	93.67	91.45	97.31	93.11	96.86	95.45	87.86	96.31	79.76	92.89
GACL	98.39	92.58	92.02	99.03	97.40	97.99	98.40	85.88	87.86	81.19	93.07
SGFTHR	98.21	**96.18**	**94.49**	97.43	93.21	96.15	96.45	89.66	90.32	80.77	93.29
FAE-Net	**99.84**	90.01	90.68	99.76	97.99	98.31	98.55	90.34	97.80	80.65	94.39
Merge-Net	99.47	90.08	90.70	**99.81**	**98.34**	97.36	**100.00**	90.70	98.60	80.60	94.56
MSFC-Net	98.77	94.90	92.35	97.58	96.30	97.02	98.88	91.20	98.10	80.83	94.59
MSCNN	98.56	94.30	90.74	99.40	98.10	98.34	98.24	88.64	**99.79**	81.10	94.72
SME-Net	99.06	95.65	91.92	98.76	98.15	**98.56**	**100.00**	**93.98**	99.06	**81.28**	**95.64**

detection network (MSFC-Net) [72]. We implement R-Libra R-CNN, GACL, FAE-Net, Merge-Net, MSCNN, SGFTHR, and MSFC-Net on our datasets, and use the same parameters as our model shown in Table IV. The compared experiments results are summarized in Tables IX and X.

The detection results in Tables IX and X show that our proposed model achieves the best mAP among all the considered methods. For the IPIU-M dataset, our method achieves the best detection results in all categories. Notably, in the NWPU VHR-10 dataset, for the small objects, such as *vehicle*, *ship*, and the intensive objects, such as *harbor*, our model gets the highest APs.

In addition, R-Libra R-CNN thinks the multilevel features extracted from the original feature pyramids are unbalanced. Therefore, it proposes a balanced feature pyramid (BFP) module to be robust against the impact of different sizes. Specifically, it normalizes the multiscale feature maps to the same resolution by *Max_pooling* and *linear interpolation*, and then calculates their mean value to end up with the mean feature map, after feature enhancement of the mean feature map, R-Libra R-CNN uses the *linear interpolation* and *Max_pooling* to recover multilevel feature maps for subsequent processing. Because R-Libra R-CNN fuses the features of multilevel feature maps by averaging, the feature uniqueness

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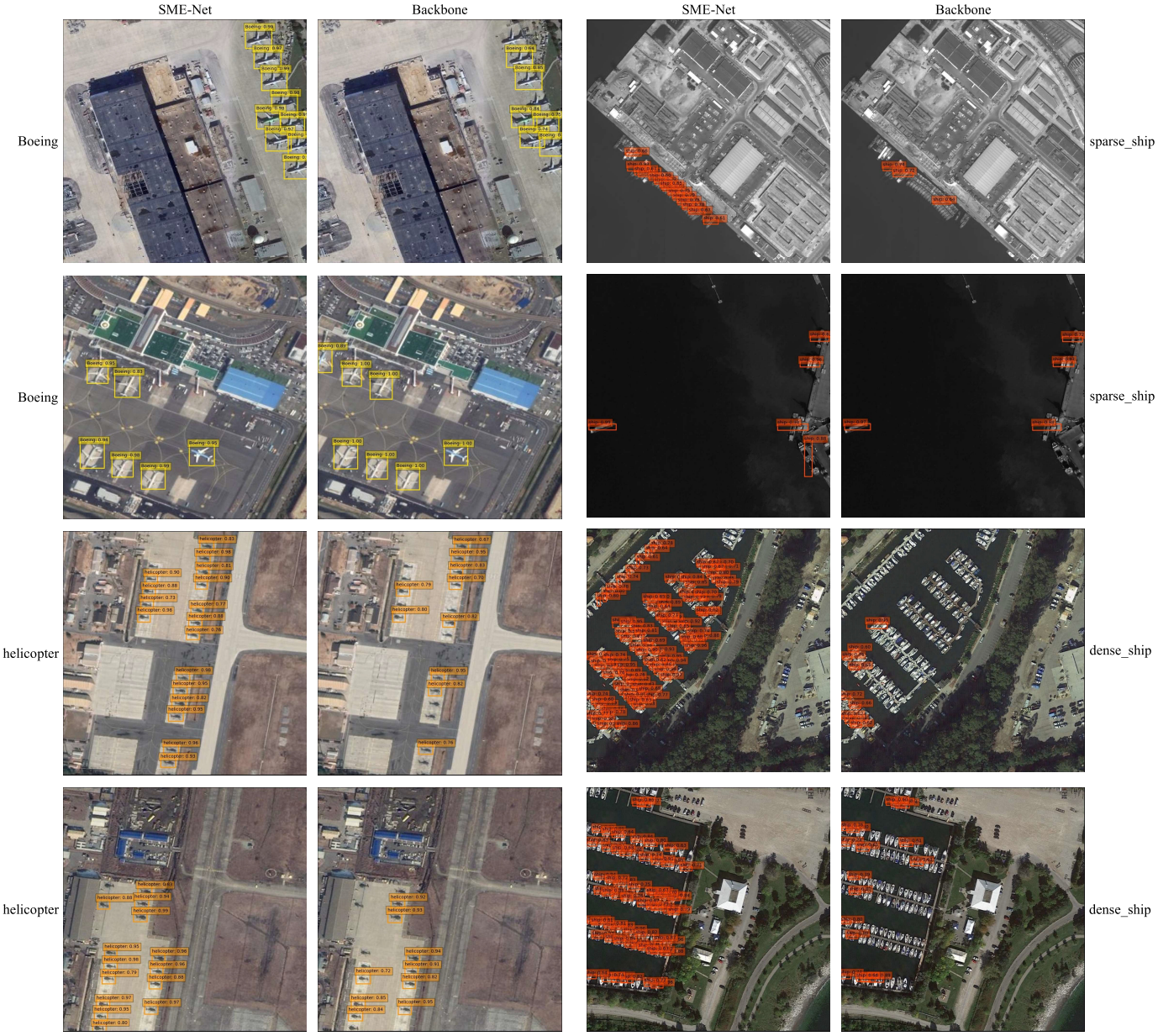


Fig. 12. Detection results comparison between the backbone network and the proposed SME-Net on IPIU-M. Numbers on those boxes are their confidence scores.

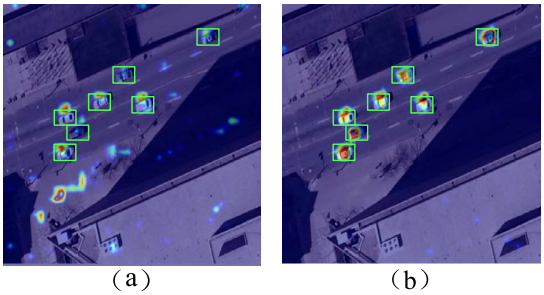


Fig. 13. Validity verification of the OSE strategy. (a) Visualization result of VGG16 U SENet. (b) Visualization result of VGG16 U OSE. The green boxes are ground truth.

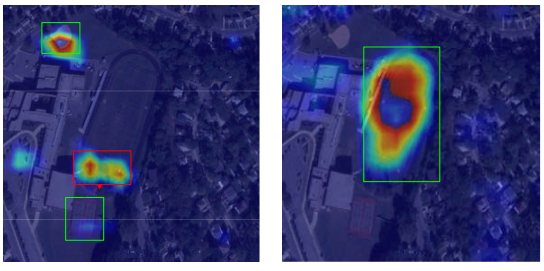
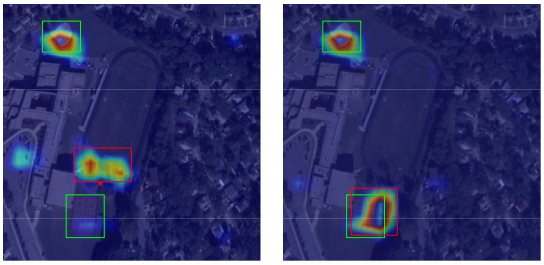


Fig. 14. Validity verification of the FSM module. The comparison of the visualized feature map (a) and (b) show that FSM splits the small and large objects. The green boxes are ground truth and the red boxes are salient region in feature map.

of the multilevel feature maps is weakened, which reduces the discriminative ability of the feature maps. Hence, its detection performance in each category is not outstanding enough.

GACL proposes an axis-concentrated prediction (ACP) module to project feature maps into different spatial directions, so as to avoid interference between coordinate axes

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(a) (b)

Fig. 15. Validity verification of the OER module. The comparison of the visualized feature map (a) and (b) show that OER reduces the deviation between strong response regions and real location. The green boxes are ground truth and the red boxes are salient region in feature map.

TABLE X
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON IPIU-M (%)

	helicopter	Boeing	dense_ship	sparse_ship	mAP
SSD-VGG16	61.89	72.89	51.37	71.14	64.32
R-Libra R-CNN	67.97	66.82	58.79	78.36	67.99
GACL	70.69	77.89	59.88	79.09	71.89
SGFTHR	69.67	76.21	58.33	82.01	71.56
FAE-Net	62.07	64.77	56.38	73.65	64.22
Merge-Net	63.50	75.20	54.90	74.36	66.99
MSFC-Net	67.67	79.20	56.37	82.31	71.39
MSCNN	69.64	73.07	57.06	63.46	65.81
SME-Net	**73.83**	**80.30**	**60.98**	**84.19**	**74.83**

and improve the location accuracy. Therefore, it achieves high detection accuracy on small objects, such as *vehicle* and *airplane*. However, it uses *Max_pooling* to concentrate information of the feature maps into two natural axes, and then obtains two axis-concentrated features p_x and p_y, which used to predict (d_x,d_w) and (d_y, d_h), respectively. Specifically, (d_x,d_y,d_h,d_w) refers to the center coordinates, height, and width of the predicted bounding box, respectively. Such operation ignores the spatial feature of the feature maps, so it is not friendly to objects with large aspect ratios, such as *bridge*.

FAE-Net applies two well-designed feature aggregation modules to adjacent feature maps in shallow layers (Conv4_3 and FC7 in SSD). Then it adds shallow feature enhancement (SFE) blocks further to strengthen the semantics of these two layers, respectively. The aggregation of information between shallow feature maps makes the texture features of the airplane more prominent, so it achieves better detection performance on the airplane. However, FAE-Net lacks the interaction between shallow and deep feature maps, and the design of the SFE module increases the depth of network, making the information of small objects easy to lose.

Merge-Net adopts a top–down connection between layers before predicting. It combines high-level features with low level-features, enabled by a deconvolutional layer and lateral connections. This strategy makes the network capture the significant detailed information in shallow layers, enhances the detailed information representation ability of deep feature maps, and further improves the detection performance of larger objects such as *baseball diamond* and *tennis court*.

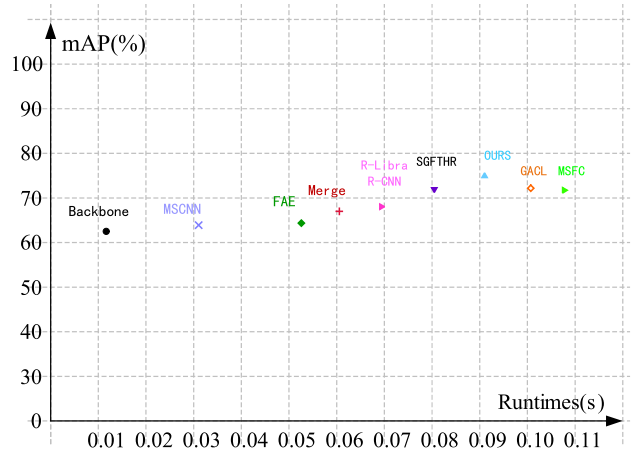


Fig. 16. Comparison of detection time with other algorithms.

MSCNN introduces dilated bottleneck structure into the backbone to improve the feature expression capacities. Dilated convolution enlarges the receptive field so that each convolution output contains a larger range of information. Especially for *bridge*, which with a large aspect ratio, the dilated convolution can extract more complete object information. Hence, MSCNN obtains better detection performance on *bridge*.

MSFC-Net proposes a compound semantic feature fusion (CSFF) module to simulate the coupled or uncoupled semantic relations semantic relationship between various pixels, thereby generating an effective semantic description. However, the CSFF module straightforwardly establishes a multilevel feature fusion from the deepest to the shallowest feature maps, and less consideration is given to the semantic deviations between all levels of the feature maps, which easily leads to nondiscriminatory information in the single-level feature maps transformed into the discriminatory information in the fusion feature maps, which has a negative impact on the feature representation ability of small objects, such as *helicopter* and *dense_ship*.

SGFTHR proposes a structure-guided feature transform (SGFT) module to prevent the essential low-level spatial and structural information from being lost when the network goes deeper. Specifically, the discriminative structural features are captured from the input image through the structure tensor extractor and guided into high-level contextual feature maps to increase informative representations. Therefore, for the small and dense objects, such as *storage tank* and *ship*, SGFTHR achieves competitive results in the dataset NWPU VHR-10. Nevertheless, in the IPIU-M dataset, most dense objects, such as *dense_ship*, are surrounded by complex backgrounds; the contrast between the objects and the background is usually not obvious. Consequently, it makes the structure information extracted by the SGFT module not accurate enough, which introduces noise into the feature map, and reduces the discriminability of features.

Throughout all categories, while improving the detection accuracy of small and dense objects, we also ensure better detection results for large-scale objects.

In addition, Fig. 16 shows the comparison of detection time between our proposed model and other models in the dataset

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TABLE XI
EXPERIMENTAL RESULTS OF EACH COMPONENT ON NWPU VHR-10 VALIDATION (%)

FSM	OER	OSE	vehicle	ship	storage tank	mAP
√			80.06	92.33	89.01	87.13
√	√		80.86	93.06	89.84	87.92
√	√	√	81.28	95.65	91.92	**89.62**

TABLE XII
EXPERIMENTAL RESULTS OF EACH COMPONENT ON IPIU-M VALIDATION (%)

FSM	OER	OSE	helicopter	Boeing	dense _ship	sparse _ship	mAP
√			69.57	77.58	56.43	73.65	69.31
√	√		71.50	78.91	57.96	74.37	70.69
√	√	√	73.83	80.30	60.98	84.19	**74.83**

IPIU-M. The average inference time of the proposed method to predict an image is by average 0.91 s. Compared with diverse methods, our algorithm needs a little more time to detect, yet, it obtains a better detection performance within the same magnitude. In the future, we will optimize our model to improve detection efficiency and ensure accuracy.

F. Ablation Study

To verify the importance of each part, we gradually add the FSM module, the OER module, and the OSE strategy to the baseline. Tables XI and XII summarize the comprehensive ablation study results. To better verify the availability of each module in our proposed method, we only describe the mAP of small and dense categories in the NWPU VHR-10 and the IPIU-M datasets.

1) FSM Module: As summarized in Tables XI and XII, the mAP increases significantly after adding the FSM module. This indicates that FSM makes the network better capture small objects' features.

2) OER Module: In Tables XI and XII, we observe that the baseline with FSM and OER outperforms the baseline only with FSM by 0.79% and 1.38% on mAP. The reason is that OER corrects the position offset in the process of spatial information transmission between adjacent feature maps and brings better regression effects to the model.

3) OSE Strategy: OSE improves the mAP from 87.92% to 89.62% and 70.69% to 74.83% in Tables XI and XII, respectively. In particular, the growth of mAP in *ship* and *storage tank* is more than 2% in the NWPU VHR-10 dataset. For small and dense objects, the feature maps extracted from the network always contain noise and blurred boundaries. OSE can effectively suppress the noise and weaken the blurred boundaries, so it effectively enhances the objects features.

V. CONCLUSION

This article proposes a feature SME-Net to improve the performance of multiscale object detection in remote sensing images. Experiments on several datasets have verified the effectiveness of our proposed modules. The FSM module alleviates feature confusion between multiscale objects. The OER module makes the model more robust to the inconsistency of the features spatial layout among multilayer feature maps. The OSE strategy enhances the salient features and suppresses the background information. However, the internal design of our module is more cumbersome, so the computational complexity is relatively high. In addition, we use horizontal bounding boxes to outline the detected objects, which are not elegant enough. In the near future, we will tailor the network while maintaining accuracy, design a simpler and more versatile module structure, and calibrate the objects more accurately, making it more suitable for remote sensing object detection.

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